



CHEMICAL IDENTITY

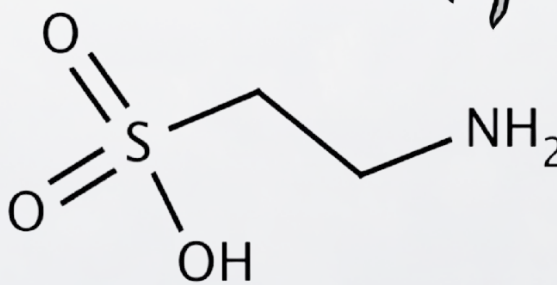
IUPAC Name: 2-Aminoethanesulfonic acid

Molecular Formula: $C_2H_7NO_3S$

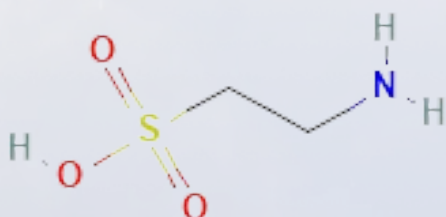
Molar Mass: 125.14 g/mol

CAS No.: 107-35-7

PubChem CID: 1123



STRUCTURE (Taurine)



INTERESTING FACT

Taurine is involved in several important physiological processes rather than serving only one function. It contributes to cell-volume regulation, membrane stabilization, calcium signaling, osmoregulation, and bile-acid conjugation.

KEY PROPERTIES & DATA

Appearance: White/colorless solid

Molecular formula: $C_2H_7NO_3S$

Molar mass: 125.14 g/mol

Functional groups: Amine + sulfonic acid

Hydrogen-bond donors: 3

Hydrogen-bond acceptors: 5

Chemical character: Sulfur-containing amino sulfonic acid

USES & VALUE

Biological role

- Helps regulate cellular water balance.
- Participates in calcium-ion regulation.
- Contributes to cell membrane stabilization.
- Helps form bile salts, which are important for lipid digestion.

Commercial/consumer use

- Studied for possible applications related to metabolism, oxidative stress, and exercise physiology.



SOURCES

Red meat, Seafood (Shells, Tuna, Salmon, Milk Fish), Supplements (Energy Drinks)



RISKS AND SAFETY

- Taurine is naturally present in the human body and in various foods.
- Research has investigated taurine's physiological and antioxidant roles.
- Evidence regarding taurine's ability to improve exercise performance is not conclusive.
- Effects can depend on the context, study design, and individual factors.

CHEMICAL ENGINEERING CONNECTION

- Taurine production involves reaction engineering to optimize reaction conditions and product yield.
- Separation and purification processes remove impurities and unreacted materials.
- Crystallization can be used to recover and purify taurine as a solid product.
- Filtration and drying separate the crystals and remove residual moisture.
- Process optimization and quality control help improve product purity, efficiency, and consistency.
- Waste treatment manages process by-products and reduces environmental impact.

